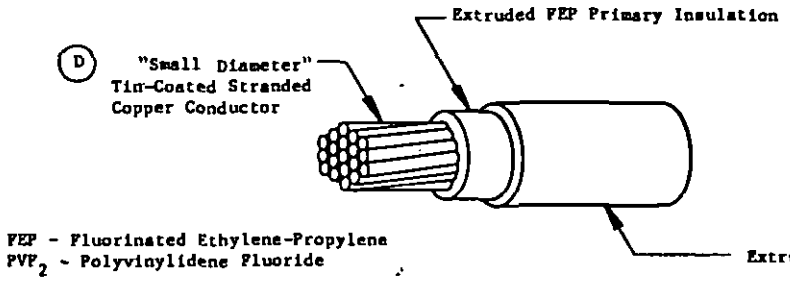
MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, FEP-PVF_2, MEDIUM WEIGHT, TIN-COATED COPPER CONDUCTOR, 600-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.



FEP - Fluorinated Ethylene-Propylene
PVF_2 - Polyvinylidene Fluoride

TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches) (D)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max) (D)	Diameter (inches)	Weight (lbs/1000 ft) (max) (D)
M22759/13-24-*	24	19 X 36	.023	.024	26.2	.049 ±.002	2.9
M22759/13-22-*	22	19 X 34	.029	.031	16.2	.051 ±.003	3.9
M22759/13-20-*	20	19 X 32	.037	.039	9.88	.061 ±.003	5.5
M22759/13-18-*	18	19 X 30	.046	.049	6.23	.071 ±.003	8.0
M22759/13-16-*	16	19 X 29	.052	.055	4.81	.077 ±.004	10.1
M22759/13-14-*	14	19 X 27	.065	.069	3.06	.091 ±.004	15.5
M22759/13-12-*	12	37 X 28	.084	.089	2.02	.111 ±.004	23.0
M22759/13-10-*	10	37 X 26	.106	.112	1.26	.138 ±.004	35.7
M22759/13-8-*	8	133 X 29	.158	.169	.701	.196 ±.006	62.8
M22759/13-6-*	6	133 X 27	.198	.212	.445	.246 ±.007	99.3
M22759/13-4-*	4	133 X 25	.250	.268	.280	.300 ±.008	153.0
M22759/13-2-*	2	665 X 30	.320	.340	.183	.374 ±.012	247.0
M22759/13-1-*	1	817 X 30	.360	.380	.149	.412 ±.014	311.0
M22759/13-01-*	0	1045 X 30	.405	.425	.116	.458 ±.016	387.0

1/ PART NO.: The asterisks in the part number column, Tables I and II, shall be replaced by color code designators in accordance with MIL-STD-681. *Examples:* Size 20, white - M22759/13-20-9; white with orange stripe - M22759/13-20-93.

(D) denotes changes

FSC 6145

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TABLE II. PERFORMANCE DETAILS

Part No.	Abrasion resistance (Procedure II)					Bend testing				
	Resistance (inches of tape) (min) (D)		Weight support bracket	Weight (lbs)	Tension load (lbs)	Mandrel diameter (inches) (±3%) (D)			Test load (lbs) (±3%) (D)	
	Initial	After immersion 1/				Life cycle (oven & bend tests) 2/	Cold bend test	Wrap test 3/	Life cycle (oven & bend tests) 2/	Cold bend test
M22759/13-24-*	20	14	A	1.0	1.0	.50	1.00	-	0.5	3.0
M22759/13-22-*	20	14	A	1.0	1.0	.75	1.00	-	1.5	3.0
M22759/13-20-*	25	17	A	1.0	1.0	.75	1.00	-	2.0	4.0
M22759/13-18-*	26	18	A	1.0	1.0	1.00	1.50	-	2.0	4.0
M22759/13-16-*	14	10	B	3.0	2.0	1.00	1.50	-	2.0	5.0
M22759/13-14-*	14	10	B	3.0	2.0	1.50	2.00	-	2.0	5.0
M22759/13-12-*	16	11	B	3.0	2.0	2.00	2.00	-	2.0	5.0
M22759/13-10-*	22	15	C	3.0	2.0	3.00	3.00	-	2.0	5.0
M22759/13-8-*	24	17	C	3.0	2.0	3.00	4.00	-	3.0	6.0
M22759/13-6-*	28	20	C	3.0	2.0	4.00	5.00	-	3.0	10.0
M22759/13-4-*	26	18	C	4.25	2.0	5.00	6.00	-	3.0	10.0
M22759/13-2-*	28	20	C	4.25	2.0	6.00	8.00	1.00	4.0	15.0
M22759/13-1-*	30	21	C	4.25	2.0	8.00	10.00	1.25	4.0	15.0
M22759/13-01-*	30	21	C	4.25	2.0	8.00	10.00	1.50	4.0	15.0

1/ Qualification test only.
2/ Also for bend tests after immersion.
3/ Use mandrel for wire sizes 2 and larger in "wrap back" test.

ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 135°C (275°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: 1500 volts (rms), 60Hz, 100% test
(D) IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), 100% test
INSULATION RESISTANCE: 5000 megohms for 1000 ft (min)
WRAP TEST: "Wrap back" test required; no cracking. (See Table II, Note 3)
Oven temperature, 160 ±2°C (320 ±3.6°F)
BLOCKING: 135 ±2°C (275 ±3.6°F)
SHRINKAGE: 0.125 inch max at 150 ±2°C (302 ±3.6°F)
WICKING: No requirement
LOW TEMPERATURE (COLD BEND):
Bend temperature: -55 ±2°C (-67 ±3.6°F)
Dielectric test, 2500 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature, 135 ±2°C (275 ±3.6°F)
Max change in measurement Sizes 24 through 12: 0.060 inch
Sizes 10 through 8: 0.100 inch
Sizes 6 through 0: 0.125
(D) FLAMMABILITY: Post-flame dielectric test not required
LIFE CYCLE: Oven temperature,160 ±2°C (320 ±3.6°F)
Dielectric test, 2500 volts (rms) 60Hz
DIELECTRIC TEST AFTER IMMERSION: 2500 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: Qualification test only
HUMIDITY RESISTANCE: 5000 megohms for 1000 ft, min insulation resistance after humidity exposure
SURFACE RESISTANCE:
Sizes 24 through 12: 500 megohm-inches (min), initial and final readings
Sizes 10 through 0: No requirement
SMOKE: 150°C (302°F)
ACID RESISTANCE: No requirement

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ADDITIONAL REQUIREMENTS (Continued)

COLOR: In accordance with MIL-STD-104, Class 1; white preferred

COLOR STRIPING OR BANDING DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min); 500 grams weight
(D) WIRE LENGTH REQUIREMENTS: Schedule A
INTENDED USE: The wire of this specification sheet is intended for interconnecting and hookup applications.

Custodians:
Navy - AS
Army - EL
Air Force - 80

Preparing activity:
Navy - AS
(Project No. 6145-0501-13)

Review activities:
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DOCUMENT IDENTIFIER AND TITLE MIL-W-22759/13D WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, FEP-PVF_2, MEDIUM WEIGHT, TIN-COATED COPPER CONDUCTOR, 600-VOLT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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